Fluctuation Formulas for Heat Capacity and Magnetic Susceptibility

Computational Physics I

Consider a system with configurations X and energy $E(X)$. In thermal equilibrium at temperature T, the probability of configuration X is the Boltzmann distribution

$$\pi(X) = \frac{e^{-\beta E(X)}}{Z}, \qquad Z = \sum_X e^{-\beta E(X)}, \qquad \beta = \frac{1}{T}, \tag{1}$$

where we set $k_B = 1$. For any observable $O(X)$,

$$\langle O \rangle = \sum_X O(X)\pi(X). \tag{2}$$

The results below are completely general: they do not depend on the dimensionality of the system or on the Ising model in particular.

1. Heat capacity from energy fluctuations

The total heat capacity is defined by

$$C = \frac{d\langle E \rangle}{dT}. \tag{3}$$

It is convenient first to differentiate with respect to β. Since

$$\langle E \rangle = \frac{1}{Z}\sum_X E(X)e^{-\beta E(X)}, \tag{4}$$

we obtain

$$\frac{\partial \langle E \rangle}{\partial \beta} = \frac{1}{Z}\sum_X \left[-E(X)^2\right] e^{-\beta E(X)} - \frac{1}{Z^2}\frac{\partial Z}{\partial \beta}\sum_X E(X)e^{-\beta E(X)}. \tag{5}$$

But

$$\frac{\partial Z}{\partial \beta} = -\sum_X E(X)e^{-\beta E(X)} = -Z\langle E \rangle, \tag{6}$$

so

$$\frac{\partial \langle E \rangle}{\partial \beta} = -\langle E^2 \rangle + \langle E \rangle^2 = -\left(\langle E^2 \rangle - \langle E \rangle^2\right). \tag{7}$$

Using

$$\frac{d\beta}{dT} = -\frac{1}{T^2} = -\beta^2, \tag{8}$$

we find

$$\boxed{C = \frac{d\langle E \rangle}{dT} = \frac{\langle E^2 \rangle - \langle E \rangle^2}{T^2} = \beta^2\left(\langle E^2 \rangle - \langle E \rangle^2\right).} \tag{9}$$

For a system of N spins, the heat capacity per spin (called c in the homework) is therefore

$$\boxed{c = \frac{C}{N} = \frac{\langle E^2 \rangle - \langle E \rangle^2}{NT^2}.} \tag{10}$$

Thus the heat capacity can be obtained directly from the *fluctuations of the energy* in an equilibrium Monte Carlo trajectory.

2. Magnetic susceptibility from magnetization fluctuations

For Ising spins $\sigma_i = \pm 1$, define the total magnetization

$$M(X) = \sum_{i=1}^{N} \sigma_i. \tag{11}$$

The energy in a magnetic field H can be written in the general form

$$E(X) = E_0(X) - HM(X), \tag{12}$$

where $E_0(X)$ contains all terms independent of H. For the one-dimensional Ising model,

$$E_0(X) = -J \sum_{i=1}^{N} \sigma_i \sigma_{i+1}, \tag{13}$$

and hence

$$E(X) = -J \sum_{i=1}^{N} \sigma_i \sigma_{i+1} - HM(X). \tag{14}$$

The magnetization density is

$$m = \frac{\langle M \rangle}{N}, \tag{15}$$

and the magnetic susceptibility per spin is defined as the response of m to the applied field,

$$\chi = \frac{\partial m}{\partial H} = \frac{1}{N} \frac{\partial \langle M \rangle}{\partial H}. \tag{16}$$

Now

$$\langle M \rangle = \frac{1}{Z} \sum_{X} M(X) e^{-\beta [E_0(X) - HM(X)]}. \tag{17}$$

Because

$$\frac{\partial}{\partial H} e^{-\beta [E_0(X) - HM(X)]} = \beta M(X) e^{-\beta [E_0(X) - HM(X)]}, \tag{18}$$

we have

$$\frac{\partial \langle M \rangle}{\partial H} = \frac{\beta}{Z} \sum_{X} M(X)^2 e^{-\beta E(X)} - \frac{1}{Z^2} \frac{\partial Z}{\partial H} \sum_{X} M(X) e^{-\beta E(X)}. \tag{19}$$

Also,

$$\frac{\partial Z}{\partial H} = \beta \sum_{X} M(X) e^{-\beta E(X)} = \beta Z \langle M \rangle. \tag{20}$$

Therefore

$$\frac{\partial \langle M \rangle}{\partial H} = \beta \left(\langle M^2 \rangle - \langle M \rangle^2 \right), \tag{21}$$

and hence

$$\boxed{\chi = \frac{\langle M^2 \rangle - \langle M \rangle^2}{NT} = \frac{\beta}{N} \left(\langle M^2 \rangle - \langle M \rangle^2 \right).} \tag{22}$$

Thus magnetic susceptibility is determined by the *fluctuations of the magnetization* in equilibrium, even when the simulation itself is performed at $H = 0$.

3. What to measure in a Monte Carlo simulation

After thermalization, an MCMC simulation produces configurations

$$X_1, X_2, \ldots, X_{N_{\text{meas}}}. \tag{23}$$

For each configuration, measure

$$E_n = E(X_n), \qquad M_n = M(X_n). \tag{24}$$

The required Monte Carlo averages are then

$$\langle E \rangle, \qquad \langle E^2 \rangle, \qquad \langle M \rangle, \qquad \langle M^2 \rangle. \tag{25}$$

From these four quantities one obtains

$$\boxed{\varepsilon = \frac{\langle E \rangle}{N}, \qquad c = \frac{\langle E^2 \rangle - \langle E \rangle^2}{NT^2}, \qquad m = \frac{\langle M \rangle}{N}, \qquad \chi = \frac{\langle M^2 \rangle - \langle M \rangle^2}{NT}.} \tag{26}$$

These are the fluctuation formulas used in the one-dimensional Ising-model homework.